

Chunk Size & the Precision–Recall Trade-off, Worked Out by Hand

Why the single most consequential RAG hyperparameter — how big you cut your documents — is a direct dial between retrieval precision and recall, quantified with an answer-density calculation and an overlap budget.

What this document does. It models a fixed-size “answer span” buried in a long document and computes how chunk size changes the *signal density* of the chunk that contains it (precision) versus the *chance the answer is split* across a boundary (recall). It then derives the overlap needed to protect boundary-straddling answers and states the practical operating points. Numbers from Appendix A.

This is an intuition PDF: the model is deliberately simplified to make the tradeoff visible; real corpora need empirical tuning (Track 6).

1 The setup

A user’s question is answered by a specific passage — say a 40-token span — sitting inside a much longer document. Your ingestion pipeline has already split that document into fixed-size chunks; retrieval will embed the question, find the most similar chunk(s), and hand them to the LLM. The chunk size S decides two things at once: how *pure* the winning chunk is, and how *likely* the answer survived chunking intact.

Intuition

A chunk is the unit of retrieval — the smallest thing you can fetch or miss. Make chunks **small** and each one is a tight, on-topic nugget (great precision: the embedding isn’t diluted), but a single idea may be sliced across two chunks and half-retrieved (recall suffers, and context is fragmented). Make chunks **large** and almost any answer fits inside one (great recall), but the chunk’s embedding is an average over many topics, so it matches weakly and drags in irrelevant text (precision suffers). There is no free size — only a tradeoff to place deliberately.

2 Precision: answer density within the chunk

Model the retrieved chunk’s “relevance density” as the fraction of its tokens that are the actual answer — a proxy for how undiluted its embedding is:

$$\text{density} = \frac{\text{answer tokens}}{\text{chunk size } S}.$$

For a 40-token answer:

chunk size S	answer density = $40/S$
128	0.312
256	0.156
512	0.078

At $S = 128$ nearly a third of the chunk is signal; at $S = 512$ the answer is under 8% of the text, the other 92% averaging into the embedding as noise. Smaller chunks $\Rightarrow$ denser signal $\Rightarrow$ sharper similarity $\Rightarrow$ higher *precision*.

3 Recall: the boundary-splitting risk

The countervailing force: a fixed grid of cut points can slice an answer in two, so neither chunk contains it whole. With chunk size S and answer length a , an answer placed uniformly at random is split whenever a cut line falls inside it. The fraction of start positions that straddle a boundary is

$$P(\text{split}) \approx \frac{a}{S} \quad (\text{for } a < S).$$

For $a = 40$: $P(\text{split}) \approx 40/128 = 0.31$ at $S = 128$, but only $40/512 = 0.08$ at $S = 512$. **Smaller chunks split answers more often**, hurting recall — the exact opposite pressure to precision. The same $40/S$ that was “good density” is also “split probability.”

Mini-example • the tradeoff is one number, read two ways

The ratio $40/S$ is the tradeoff. Shrinking S raises it — which simultaneously means “more signal per chunk” (precision $\uparrow$) and “more likely to cut the answer” (recall $\downarrow$). You cannot move one without the other. That is why chunk size is a *position* you choose on a curve, not a value you optimise to zero error.

4 Overlap: buying back boundary recall

Overlap is the standard fix for splitting: let consecutive chunks share their last/first O tokens, so an answer cut by one grid line is still whole inside the neighbour.

$$\text{stride} = S - O, \quad \text{an answer of length } a \leq O \text{ is } \textit{never} \text{ orphaned.}$$

Worked budget. $S = 512$, overlap $O = 64 \Rightarrow$ stride = 448. Any answer up to 64 tokens that lands near a boundary appears intact in the overlapping chunk, so boundary-splitting recall loss is eliminated for $a \leq 64$ — and our 40-token answer is safely covered. The cost is redundancy: with stride 448 vs size 512, you store and search $512/448 = 1.14\times$ more chunks ($\sim 14\%$ overhead).

Intuition

Overlap lets you keep larger-chunk recall *and* reduce splitting without shrinking S — at a storage/compute tax of $S/(S-O)$. Set O to roughly your typical answer or sentence length: big enough to span boundaries, small enough that the overhead stays modest. Common defaults ($S \approx 256\text{--}512$, $O \approx 10\text{--}20\%$) are just sensible points on these two curves.

5 Choosing the operating point

- **Small (~ 128):** precise, great for fact lookup / short answers; risks fragmenting multi-sentence reasoning. Use with overlap and/or a re-ranker.
- **Large ($\sim 512+$):** high recall, preserves context for synthesis questions; dilutes embeddings and feeds the LLM more noise per chunk.
- **Structure-aware:** cut on natural boundaries (paragraphs, headings, sentences) instead of a blind token grid — this attacks the splitting term directly, often beating any fixed size.

The honest answer is empirical: sweep size and overlap, measure retrieval recall (#12) and end-to-end answer quality (Track 6), and place the dial where *your* questions and documents want it.

Equation cheat-sheet

$$\text{density} = \frac{a}{S} \text{ (precision } \uparrow \text{ as } S \downarrow), \quad P(\text{split}) \approx \frac{a}{S} \text{ (recall } \downarrow \text{ as } S \downarrow)$$

$$\text{stride} = S - O, \quad a \leq O \Rightarrow \text{no boundary loss}, \quad \text{overhead} = \frac{S}{S - O}$$

Appendix A · Reference sketch

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a = 40                                # answer length in tokens
for S in [128, 256, 512]:
    print(S, round(a / S, 3))         # density AND approx split-prob
# 128 -> 0.312    256 -> 0.156    512 -> 0.078

S, O = 512, 64
stride = S - O                        # 448
overhead = S / (S - O)                # 1.14x storage/compute
# answers up to O=64 tokens near a boundary survive in the overlap chunk

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Internal learning material. Numbers reproducible from the appendix. v1.0